

# EECS 841

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article amsmath geometry margin=1in

## Output Dimension Calculation

For a convolution operation with:

- Input size:  $3 \times 4$
- Filter size:  $2 \times 2$
- Padding: 0
- Stride: 1

### Output Height:

$$\text{Output Height} = \left\lfloor \frac{3 - 2 + 0}{1} \right\rfloor + 1 = \lfloor 1 \rfloor + 1 = 2$$

### Output Width:

$$\text{Output Width} = \left\lfloor \frac{4 - 2 + 0}{1} \right\rfloor + 1 = \lfloor 2 \rfloor + 1 = 3$$

**Final Output Size:**  $2 \times 3$